REWEAR-FUSED: The Absolute Ceiling Backend Architecture for Co-Designed Recyclable Stretch Apparel (2026 Frontier Spec)

TL;DR

- The maximum-ceiling REWEAR-FUSED backend is an agentic, all-atom, closed-loop co-design platform built around **RFdiffusion3** (atom-level diffusion, Dec 3, 2025 open-source release, 168M parameters, ~10× faster than RFdiffusion2) + **LigandMPNN / EnhancedMPNN** (catalysis-aware inverse folding; ResiDPO triples enzyme design success rate from 6.56% → 17.57%) + **Boltz-2 / BoltzGen** (joint structure + FEP-accurate affinity, Pearson R = 0.62 on full 876-pair OpenFE benchmark, 1,000× faster than FEP) + **PLACER** (catalytic ensemble validation, PNAS Nov 4, 2025) + **MACE-OFF24 / OrbMol / UMA-M** universal ML potentials for QM-accurate MD, orchestrated by an LLM "AI co-scientist" (ProtAgents/AgentD/Robin-style) on a Nextflow/Flyte graph spanning H200, B200, and GB200 NVL72 racks — all licensing-safe (MIT/CC-BY) and explicitly avoiding AlphaFold3/ESM3/Chai-2.
- For Engine B specifically, the frontier is to *not* copy UMG-SP2 but to use its QM/MM-validated Ser-Sercis-Lys transition state (Gibbs activation energy 20.8 kcal/mol per Polêto et al., RSC Chem. Sci. 2025, DOI 10.1039/d4sc06688j) as a *theozyme* prompt to RFdiffusion2/3, score with PLACER over the full reaction coordinate (acyl-enzyme intermediate included), refine sequence with ResiDPO-trained EnhancedMPNN, then validate with Boltz-2 absolute binding free energy (Boltz-ABFE pipeline, Recursion, Oct 2025) and MACE-OFF24 QM/MM-grade dynamics — a stack that already delivers cell-comparable catalytic proficiency in silico (RFdiffusion2 solved 41/41 enzyme benchmarks vs 16 for the previous best, per IPD/Nature Methods, Dec 3, 2025).
- The "shouldn't-be-possible" ceiling is a tri-engine, agentic, multi-objective Bayesian-optimization loop (Atlas + Honegumi + ChemOS 2.0) in which the fiber (Engine A), enzyme (Engine B), and Digital Product Passport (Engine C) co-evolve every iteration — fiber chemistry mutates → urethanase theozyme re-prompts RFdiffusion3 → Boltz-2 re-scores binding → MACE-OFF24 verifies cleavage kinetics → ESPR passport regenerates — driven by Gemini 2.0/Claude-class LLM agents using the AI Scientist v2/Kosmos paradigm (Kosmos from Edison Scientific processes ~1,500 papers + 42,000 lines of analysis code per run, $70M seed Dec 18, 2025). IntuitionLabs

Key Findings

Engine B is now a solved problem at proof-of-concept — and the frontier is to push past it

The Baker lab in December 2025 announced that **RFdiffusion2** delivers catalytic proficiencies *comparable to enzymes found in cells* and solved **41 of 41** difficult enzyme-design challenges vs only 16 for the previous best tool (Ahern, Yim, Tischer, Woodbury et al., Nature Methods 2025, DOI 10.1038/s41592-025-02975-x, code github.com/RosettaCommons/RFdiffusion2). The companion metallohydrolase paper (Kim, Woodbury, Tischer et al., Nature 2025, DOI 10.1038/s41586-025-09746-w) reached catalytic efficiencies up to 23,000 $M^{-1}s^{-1}$. **RFdiffusion3** (Butcher, Krishna et al., bioRxiv 10.1101/2025.09.18.676967, open-sourced Dec 3, 2025) is a 168M-parameter atom-level diffusion model that is ~10× faster than RFdiffusion2; it screened **190 cysteine-hydrolase designs to find 35 multi-turnover catalysts with top kcat/Km = 3,557 $M^{-1}s^{-1}$**. Combined with **Lauko et al., Science 388:eadu2454, April 18, 2025** — which built de novo serine hydrolases with kcat/Km up to 2.2×10^5 $M^{-1}s^{-1}$ on five non-natural folds — the recipe for designing a brand-new carbamate-hydrolase anchored on the UMG-SP2 transition state is fully buildable today. uw biorxiv

UMG-SP2 is the perfect anchor — and the frontier is to *transcend* it

Polêto et al. (RSC Chemical Science 2025, DOI 10.1039/d4sc06688j) published a complete QM/MM mechanism for UMG-SP2 urethanase: Ser-Sercis-Lys triad, two stages (acylation + deacylation), rate-limiting Gibbs activation energy 20.8 kcal/mol, with specific point mutations identified that stabilize the rate-limiting TS macrodipole. Gan/Liu et al. (Adv. Sci. 2025, PMC11967865) crystallized UMG-SP2 (PDB 8WDW, 2.59 Å) and engineered variants with **>30.7-fold higher PU depolymerization activity**. The REWEAR-FUSED ceiling is to feed the *transition-state coordinates* (not the protein) into RFdiffusion2/3 as a theozyme prompt — exactly the workflow proven in Ahern et al. — and produce a Ser-His-Asp triad (or, if Engine A's substrate permits, a native Ser-Sercis-Lys) that catalyzes the same carbamate hydrolysis at near-neutral pH on a novel scaffold with no UMG homology (FoldSeek TM-score < 0.5). nih PubMed Central

Boltz-2 is the affinity oracle — and Boltz-ABFE closes the loop with physics

Boltz-2 (Passaro et al., MIT/Recursion, June 2025, bioRxiv 10.1101/2025.06.14.659707, MIT license) is the first deep-learning model to approach FEP accuracy: Pearson R = 0.66 on the 4-target FEP+ subset, R = 0.62 on the full 876-pair OpenFE benchmark — comparable to OpenFE while being **1,000× faster** (~$0.001/prediction in ~20 s on one GPU vs ~$100/6–12 h for FEP). It topped the **CASP16** blind affinity challenge across 140 protein-ligand pairs. Recursion's **Boltz-ABFE pipeline** (Bio-IT World, Oct 10, 2025) couples Boltz-2 to absolute-binding-free-energy simulations, creating a closed loop where 8 of 10 (HLL library) and 10 of 10 (kinase library) top-ranked compounds were predicted as binders with |R| = 0.74 ABFE/screen-score correlation. For Engine B, this means every designed urethanase candidate gets joint structure + affinity in ~20 s on a single H100, with optional Boltz-ABFE physics verification. Pubs

LigandMPNN → EnhancedMPNN/ResiDPO is the frontier of catalysis-aware sequence design

LigandMPNN (Dauparas et al., Nature Methods 2025, DOI 10.1038/s41592-025-02626-1) outperforms Rosetta and ProteinMPNN on sequence recovery for ligand-binding (63.3% vs 50.4%/50.5%), nucleotide (50.5% vs 35.2%/34.0%), and metal-binding (77.5% vs 36.0%/40.6%) residues — and has 100+ experimentally validated designs. The frontier extension is **EnhancedMPNN** via **Residue-level Designability Preference Optimization (ResiDPO)** (Xue et al., arXiv 2506.00297, May 30, 2025), which fine-tunes LigandMPNN against AlphaFold pLDDT and delivers a **near 3-fold increase in enzyme in-silico design success (6.56% → 17.57%) and a 2.3× boost in binder success** at the same ~55% sequence recovery. For multi-state design across the catalytic cycle (Michaelis complex, acyl-enzyme intermediate, deacylation TS), LigandMPNN + FastRelax + PLACER over each state is the proven Lauko et al. recipe — push it further by replacing LigandMPNN with EnhancedMPNN.

The all-atom binder frontier expands the toolset: BoltzGen and Chai-2

BoltzGen (Stärk et al., MIT/Boltz/NVIDIA et al., bioRxiv 10.1101/2025.11.20.689494, released MIT-licensed Oct 26, 2025, code at github.com/HannesStark/boltzgen) is an all-atom generative model that unifies design + structure prediction, achieving **nanomolar binders for 66% (6/9) of novel targets** with <30% sequence identity to any bound PDB protein, testing 15 designs per target. **Chai-2** (Chai Discovery, June 2025, bioRxiv 10.1101/2025.07.05.663018) achieves a **16% hit rate in fully de novo antibody design** (a ~100× improvement over prior methods) and **68% miniprotein success with picomolar affinities** — but is proprietary/early-access and therefore licensing-incompatible. For REWEAR-FUSED's commercial-safe frontier, BoltzGen + RFdiffusion3 is the open-source pair. biorxiv

Machine-learning interatomic potentials hit the QM-accuracy/MD-speed crossover in 2025

- **MACE-OFF24/25 (medium)** (Kovács et al., JACS 147:17598–17611, 2025, DOI 10.1021/jacs.4c07099; arXiv:2312.15211 v5 Aug 2025) — 6 Å-cutoff transferable organic ML force field validated on solvated proteins and peptide folding: "We obtained a throughput of 3×10^5 steps per day on a single NVIDIA A100 80GB GPU." This is the relevant tool for urethanase + polymer substrate MD with near-QM fidelity. PubMed Central
- **MACELES-OFF** (Cheng et al., arXiv:2507.14302, Jul 18, 2025) — adds long-range electrostatics via Latent Ewald Summation to MACE-OFF; critical for charged Ser/Lys/Asp/His triads and bulk liquid water.
- **UMA** (Universal Models for Atoms, Wood et al., Meta FAIR + CMU, arXiv:2506.23971, Jul 1, 2025) — trained on ~500M unique 3D atomic structures; UMA-S/M/L family with mixture-of-linear-experts; UMA-L is 700M parameters supporting >1k atoms. CC-licensed via Hugging Face. arXiv arxiv

- **OrbMol** (Orbital Materials, Aug 2025, orbitalmaterials.com/posts/orbmol-extending-orb-to-molecular-systems) — extends Orb-v3 to molecules using Open Molecules 2025 (OMol25)'s >100M ωB97M-V/def2-TZVPD DFT calculations; explicitly supports metal complexes, biomolecules, electrolytes. **Orb-v3** (Rhodes et al., arXiv:2504.06231, Apr 8, 2025) achieves >10× latency and >8× memory reduction vs Orb-v2. Orbitalmaterials arXiv
- **AIMNet2** (Anstine, Zubatyuk, Isayev, Chem. Sci. 16:10228–10244, Apr 29, 2025, DOI 10.1039/d4sc08572h) handles 14 elements including charged/radical organics; **AIMNet2-NSE** (Kalita et al., Angew. Chem. Int. Ed. 65:e16763, Dec 16, 2025, DOI 10.1002/anie.202516763) adds spin-charge equilibration for open-shell radical chemistry — directly relevant to polyurethane curing chemistry. nih
- **SevenNet-Omni** (Kim et al., arXiv 2510.11241, Oct 2025) trained on 15 open databases and 250M structures. arxiv
- **MACE-MP-0b/0b2/0b3 + MACE-MPA-0** (ACEsuit/mace-foundations, 2025 rolling releases) — state-of-the-art on Matbench Discovery for inorganic.

Polymer informatics has its own foundation models — and the frontier is BART-style generative + multimodal

- **polyBERT** (Kuenneth & Ramprasad, Nat. Commun. 14, 2023; widely cited 2025) — 100M-PSMILES Transformer fingerprint trained on 100M hypothetical PSMILES across 29 properties; two-orders-of-magnitude speed-up over classical fingerprints + random forests.
- **polyBART** (Agarwal et al., arXiv 2506.04233, Jun 2025) — the *first* polymer foundation language model with bidirectional structure↔property generation, trained on Pseudo-polymer SELFIES (PSELFIES); reports an experimentally synthesized and validated polymer designed by a language model (predicted to exhibit high thermal degradation temperature, confirmed by lab measurement).
- **NeurIPS 2025 Open Polymer Prediction Challenge (2,241 teams)** (per arXiv 2511.10893 from the 9th-place team) — the winning solution by James Day showed that **ModernBERT-base outperformed both ChemBERTa and polyBERT** with property-specific ensembles and LAMMPS-derived 3D molecular descriptors for the five target properties (Tg, Tc, density, FFV, Rg). Key insight: REWEAR-FUSED Engine A should not rely on a single polymer model — use an ensemble.
- **TransPolymer** + **MoLFormer-XL** (Ross et al., IBM, 1.1B SMILES from PubChem+ZINC, masked language modeling, linear attention with RoPE) remain in active use; ChemBERTa-3 (Chemrxiv Aug 2025) provides a standardized open-source benchmarking framework. Hunter Heidenreich + 2
- For mechanical/elastomer properties, LAMMPS-driven coarse-grained MD remains the validation backbone (used in NeurIPS winner's stack).

Hardware: the GB200 NVL72 is the new ceiling

NVIDIA Blackwell shipped in late 2025:

- **B200**: 180–192 GB HBM3e, 8 TB/s memory bandwidth, 9,000 TFLOPS FP4, NVLink 5.0 @ 1.8 TB/s bidirectional. ~3–4× H100 training, 11–15× LLM inference per Exxact Corp/SemiAnalysis benchmarks; available on RunPod at $4.99/hr on-demand. Runpod + 6
- **GB200 Superchip**: 2× B200 + 1× Grace ARM CPU via NVLink-C2C @ 900 GB/s; 384 GB combined HBM3e at 16 TB/s. NVIDIA Developer CUDO Compute
- **GB200 NVL72**: 72× B200 + 36× Grace CPUs in a single rack delivering **720 PFLOPS of training (FP8) and 1.44 exaFLOPS of FP4 inference** per NVIDIA VP Ian Buck (DataCenter Dynamics GTC briefing, mirrored on nvidia.com/en-us/data-center/gb200-nvl72/); NVIDIA's official product page advertises "30× faster real-time trillion-parameter LLM inference" vs H100 generation (benchmarked at TTL=50ms, FTL=5s, 32,768-token input / 1,024-token output throughput). Jarvis Labs + 4
- **H200**: 141 GB HBM3e, 4.8 TB/s — the workhorse for RFdiffusion2/3 and Boltz-2 inference. RunPod lists H200 at $4.39/GPU-hour on-demand as of May 2026 (runpod.io/gpu-models/h200). Runpod SaaS Sentinel

Orchestration: Nextflow/nf-core is the proven backbone, Atlas/ChemOS 2.0 is the SDL brain

- **nf-core/proteinfold** (active 2025) integrates AlphaFold2, ColabFold, ESMFold with multi-model 3D visualization (BioCommons Australia 2025 release).
- **seqeralabs/nf-proteindesign** (late 2025, github.com/seqeralabs/nf-proteindesign) integrates **BoltzGen + ProteinMPNN + Boltz-2 refold + IPSAE scoring** for protein, nanobody, and peptide binder design.
- **nf-core/proteinfamilies** (Karatzas et al., GigaScience 15:giag009, 2026, DOI 10.1093/gigascience/giag009) demonstrates linear-scaling protein-family pipelines on AWS.
- **Atlas** (Hickman et al., Digital Discovery 4:1006–1029, 2025, DOI 10.1039/D4DD00115J, code at github.com/aspuru-guzik-group/atlas) is the "brain" for self-driving labs, integrating Phoenics + Gryffin + Olympus + Golem on top of Ax/BoTorch. ScienceDirect
- **ChemOS 2.0** (Sim, Vakili, Strieth-Kalthoff et al., Matter, 2024) is the SiLA2-based orchestration OS for chemical SDLs.
- **Honegumi** (Baird et al., arXiv 2502.06815, Feb 4, 2025, honegumi.readthedocs.io) generates Ax/BoTorch BO scripts interactively for non-expert scientists.
- **NIMO** (formerly NIMS-OS; Tandfonline DOI 10.1080/27660400.2025.2565144, Nov 4, 2025) integrates non-Python legacy robotics into AI-driven SDLs. Taylor & Francis Online
- **Bluesky** (NSLS-II, arXiv 2509.22959, Sep 2025) — modular open-source framework for collaborative human-AI multi-modal, multi-beamline synchrotron experiments. arxiv

Agentic LLM molecular design is now real

- **Coscientist** (Boiko, MacKnight, Kline, Gomes, Nature 624:570–578, Dec 20, 2023, DOI 10.1038/s41586-023-06792-0) — GPT-4-driven autonomous chemistry agent that successfully optimized Pd-catalyzed cross-couplings across six tasks. Nature Ctisus
- **ChemCrow** (Bran et al., Nat. Mach. Intell. 6:525–535, May 8, 2024, DOI 10.1038/s42256-024-00832-8; arXiv 2304.05376) — GPT-4 + 18 expert tools; autonomously planned and executed syntheses of an insect repellent and three organocatalysts and guided discovery of a novel chromophore.
- **AI Scientist-v2** (Yamada/Lange et al., Sakana AI, arXiv 2504.08066, Apr 10, 2025; code github.com/SakanaAI/AI-Scientist-v2) — first AI-generated paper to pass workshop peer review at ICLR via agentic tree search. ResearchGate
- **Google AI co-scientist** (Gottweis & Natarajan, research.google blog, Feb 19, 2025) — Gemini-2.0 multi-agent hypothesis generator. Google Research
- **FutureHouse Robin** (Hinks, Ghareeb, Chang, Mitchener, arXiv 2505.13400, May 20, 2025; github.com/Future-House/robin) — Crow/Falcon/Finch trio that autonomously discovered ripasudil as a candidate for dry AMD with 7.5-fold RPE phagocytosis enhancement. Futurehouse + 2
- **PaperQA2** (FutureHouse) — superhuman scientific-literature QA (12.4% above closest competitor on RAG-QA-Arena's science benchmark; switched to CalVer in Dec 2025). Futurehouse GitHub
- **Kosmos** (Edison Scientific, FutureHouse spinout, Nov 2025) — processes ~1,500 papers + 42,000 lines of analysis code per run. **$70M seed at ~$250M valuation, Dec 18, 2025, co-led by Triatomic Capital, Spark Capital, and a major US institutional biotech investor**; angels include Jeff Dean (Google chief scientist) and Dmitri Alperovitch (CrowdStrike co-founder). Tech Funding News + 3
- **AgentD** (Ock, Sharma Meda, Badrinarayanan et al., arXiv 2507.02925, Jun 26, 2025; J. Chem. Inf. Model., DOI 10.1021/acs.jcim.5c02454, Dec 12, 2025; code github.com/hoon-ock/AgentD) — drug-discovery LLM agent that orchestrates **Boltz-2** as an external tool to produce 3D protein-ligand complexes with IC50 + inhibitor probability. bioRxiv
- **ProtAgents** (Ghafarollahi & Buehler, Digital Discovery 3:1389–1409, May 17, 2024, DOI 10.1039/d4dd00013g) — multi-agent LLM platform for de novo protein design combining physics and ML. PubMed Central
- **Cradle Bio** won Adaptyv Bio's EGFR competition with a **1.21 nM binder, 8.2× better than Cetuximab scFv (9.94 nM)** and 4× better than the nearest competitor (5.18 nM); 130 teams submitted 1,131 designs, 400 tested by SPR, only 2 outperformed Cetuximab (cradle.bio/blog/adaptyv-protein-design-competition; bioRxiv 10.1101/2025.04.17.648362). **Tamarind Bio** provides hosted AlphaFold/RFdiffusion/MPNN/GROMACS GPU orchestration via API/web UI. Cradle + 3

Details

PART 1 — De Novo Enzyme Design at the Absolute Frontier (Engine B)

(a) Backbone generation — beyond baseline RFdiffusion. The single most important upgrade from "RFdiffusion baseline" to the 2026 ceiling is to use **RFdiffusion2** (Ahern et al., bioRxiv 10.1101/2025.04.09.648075, Nature Methods 2025, DOI 10.1038/s41592-025-02975-x) for theozyme-anchored design and **RFdiffusion3** (Butcher/Krishna et al., bioRxiv 10.1101/2025.09.18.676967 v2, open-sourced Dec 3, 2025) when ligand/substrate co-design is required. RFdiffusion2 introduced atom-level active-site scaffolding using flow matching: it removes the need for sidechain rotamer construction and sequence-position prespecification. RFdiffusion3 introduced full atom-level diffusion with 14 atoms per residue (backbone + sidechains explicit) and supports atom-level conditioning on hydrogen-bond donors/acceptors, burial states, and ligand identities — "We thus shrunk the Pairformer from 48 to just 2 layers, greatly reducing computational cost, and ultimately yielding only 168M trainable parameters for the final network (vs. ~350M for AF3)… achieving approximately a 10-fold increase in speed over RFdiffusion2" (RFD3 bioRxiv v2). For the carbamate-hydrolase task, the ceiling recipe is: (i) compute the theozyme from the UMG-SP2 Polêto et al. QM/MM-optimized transition state (TS5), (ii) submit it as an atom-level constraint set to RFdiffusion2, (iii) for any designs requiring the polyurethane substrate to be co-generated, escalate to RFdiffusion3. biorxiv

Alternative backbone generators benchmarked in 2025: **Chroma** (Ingraham et al., Generate Biomedicines, Nature 623:1070, 2023) with ChromaDesign; **Genie 2** (Lin et al., arXiv 2405.15489, 2024) which sets the SoTA for the full structural universe up to 500 residues; **Proteus** (Wang et al., bioRxiv 10.1101/2024.02.10.579791) which beats RFdiffusion on designability and matches Chroma on speed without pretraining; **FrameFlow / FoldFlow** (Yim et al. 2023; Bose et al. 2024) flow-matching variants; **Proteína** (Geffner et al. 2025) Cα-only sampling. For pure novelty/scaffolding the ceiling stack is RFdiffusion3 + Genie 2 + Proteus running in parallel and **FoldSeek**-filtering to ensure scaffolds resemble nothing in nature (TM-score < 0.5 to UMG-SP2 and zero similar PDB hits).

Family-wide hallucination / motif scaffolding for Engine B. The Lauko et al. (Science 388:eadu2454, April 18, 2025) "computational design of serine hydrolases" recipe is the gold standard and explicitly maps onto REWEAR-FUSED's locked stack: theozyme of catalytic triad + oxyanion hole → RFdiffusion1 backbones → LigandMPNN sequences → multi-state PLACER assessment across Michaelis complex, acyl-enzyme intermediate, and deacylation TS. They achieved kcat/Km up to **2.2×10^5 $M^{-1}s^{-1}$** (the "momi120_103" variant) on five non-natural folds, with crystal structures matching design models within Cα RMSD < 1 Å (PDB IDs 9DED, 9DEE, 9DEF, 9DEG, 9DEH, 9MRB). Bump from RFdiffusion1 to RFdiffusion2 to capture additional success rate.

(b) Sequence design — beyond baseline LigandMPNN. The locked stack already specifies LigandMPNN, which is correct: it outperforms ProteinMPNN/Rosetta by 13–40 percentage points on ligand/nucleotide/metal sequence recovery. Push to the ceiling with **EnhancedMPNN** (Xue et al., arXiv 2506.00297, May 30, 2025), which uses ResiDPO to triple enzyme design success rate (6.56% → 17.57%) at the same ~55% sequence recovery and pLDDT-accuracy of 66.08% (up from 57.71% for LigandMPNN and 62.11% for standard DPO). For inverse folding with explicit nucleotide/ion context, **ADFLIP** (Khan et al., arXiv 2507.14156, Jul 2025) is the discrete-flow-matching alternative achieving RMSD < 1.3 Å on small-molecule ligand contexts. The frontier extension is to design over **multiple catalytic states simultaneously**: bind LigandMPNN/EnhancedMPNN with FastRelax across (i) apo, (ii) Michaelis complex, (iii) tetrahedral TS, (iv) acyl-enzyme intermediate, (v) deacylation TS — exactly the Lauko et al. recipe that achieved multi-turnover catalysis. For REWEAR-FUSED's near-neutral-pH Ser-His-Asp triad on a carbamate substrate, additional context residues include the oxyanion-hole backbone amides (~2.8 Å N-H···O=C) and a histidine-stabilizing catalytic acid.

(c) Structure + affinity + function prediction — Boltz-2 / BoltzGen / Chai-2. Boltz-2 is the ceiling for licensing-safe co-folding (MIT license, 1,000× faster than FEP, Pearson R = 0.66/0.62 on FEP+ benchmarks, top of CASP16). Couple with **Boltz-ABFE** for any candidate that needs absolute binding free energy: 8/10 (HLL) and 10/10 (kinase) top-10 compounds predicted as binders, |R| = 0.74 ABFE/screen correlation. **BoltzGen** is the licensing-safe analog of Chai-2 for protein/peptide binder generation. **Chai-2** is excluded (proprietary, early-access only). **AlphaFold3** is excluded (Google DeepMind non-commercial license). Note Boltz-2's known limitations from independent benchmarks: Lukauskis et al. (Ternary Therapeutics) found "generally poor or even negative correlations" on a 93-compound molecular-glue ternary set; Rognan et al. found Boltz-2 *outperformed* other methods on ULVSH binder/non-binder discrimination (per Rowan Sci tracking dashboard). The frontier protocol is to use Boltz-2 for fast ranking and reserve **OpenFE / Boltz-ABFE** for the top ~100 candidates. Rowan Documentation

(d) Catalytic geometry validation — PLACER and successors. PLACER (Anishchenko, Kipnis, Kalvet et al., PNAS 122:e2427161122, Nov 4, 2025, DOI 10.1073/pnas.2427161122; bioRxiv 10.1101/2024.09.25.614868; code github.com/baker-laboratory/PLACER) is a graph neural network trained on **226,684 CSD organic small molecules + 112,828 PDB protein-ligand complexes** that recapitulates atomic positions from partially corrupted structures via stochastic denoising (Gaussian noise σ = 1.5 Å). It enables conformational ensembles for protein-small molecule pockets with up to 600 heavy atoms in a crop. For REWEAR-FUSED, PLACER runs on each Engine B candidate over each catalytic state to filter for "preorganized" active sites — Lauko et al. showed this substantially increases experimental success rates. Successors include the broader **ChemNet** that powered Lauko et al.

(e) QM/MM validation of catalysis. For each top design from PLACER, run **GROMACS 2025/2026 + CP2K** QM/MM (native CP2K interface using GEEP electrostatic embedding) or **GENESIS SPDYN + QSimulate-QM** (Yagi et al., PMC12020374, 2025; ~1 ns/day DFTB, ~10 ps/day DFT on a single node). For umbrella-sampling free-energy profiles of the carbamate-cleavage TS, target the Polêto et al. (RSC Chem. Sci. 2025) protocol: define the C-O bond-breaking + nucleophilic attack coordinates, run US, expect a designed activation energy ⩽ 20.8 kcal/mol (Lauko-style multi-turnover designs achieve substantially lower equivalents). For semi-empirical pre-screening before QM/MM, use **MACE-OFF24(M)** organic MLIP — JACS 2025 confirmed solvated-protein NPT stability at 3×10^5 steps/day on a single A100 80GB. **AIMNet2-NSE** (Dec 16, 2025) handles charged/radical intermediates for the tetrahedral intermediate states. PubMed Central

(f) Generative end-to-end frontier. BoltzGen is the closest licensing-safe end-to-end generator that jointly designs sequence + structure + binding for arbitrary modalities; nanomolar binders for 66% of novel targets in 8 wet-lab campaigns across 26 targets. The frontier-of-the-frontier for enzymes specifically (vs binders) is to chain RFdiffusion3 (backbone + ligand co-generation) → EnhancedMPNN (catalysis-aware sequence) → BoltzGen-style folding check → PLACER (catalytic geometry) → Boltz-2 affinity → MACE-OFF24 QM/MM kinetic check — wrapped in an LLM agent that iterates. bioRxiv nih

PART 2 — MD and Physics-Based Validation at the Frontier

(a) MD stacks. GROMACS 2025/2026, OpenMM 8+, and AMBER 24 remain the backbones. The 2026 frontier is to plug in universal MLIPs for near-QM accuracy at MD speed. The best fit for an enzyme-substrate complex is **MACE-OFF24(M)** for the organic protein+substrate region, optionally with **MACELES-OFF** for long-range electrostatics. For metalloenzyme variants (if Engine A's substrate prefers Zn^{2+}-activated cleavage), **UMA-M** or **OrbMol** (built on Orb-v3, trained on OMol25's 100M ωB97M-V DFT, supports metal complexes + biomolecules + electrolytes) are the ceiling. **AIMNet2-NSE** handles open-shell radical chemistry. All are MIT/CC-licensed.

(b) Free-energy methods. For binding affinity: **OpenFE** (open-source relative FEP) as the gold standard, **Boltz-ABFE** (Recursion) for absolute binding free energy at scale, **FEP+** (Schrödinger) for reference. For catalysis: **umbrella sampling** (US) over reaction coordinates (Polêto et al.

style), **metadynamics** (PLUMED 2.10+ in GROMACS/OpenMM), **string method** for minimum free-energy paths, **adaptive biasing force (ABF)** for cleavage transitions.

(c) Enhanced sampling for the full catalytic cycle. Replica-exchange with solute tempering (REST2), Hamiltonian-replica exchange, well-tempered metadynamics across the carbamate-hydrolysis reaction coordinate (C-N bond breaking + tetrahedral intermediate formation + alcohol leaving + carbamic-acid release). For polyurethane chain dynamics, **MARTINI 3** coarse-grained MD for the elastomer matrix.

(d) Validating substrate binding + TS stabilization for designed enzyme on polyurethane-derived carbamate. Protocol: (i) build the Engine A polymer fragment with the target carbamate; (ii) dock to designed enzyme via PLACER + Boltz-2; (iii) absolute binding free energy via Boltz-ABFE; (iv) QM/MM US in GROMACS 2025-CP2K (or GENESIS-QSimulate) for the TS — target $\Delta G^\ddagger \leqslant 18$ kcal/mol; (v) MACE-OFF24 free-MD for stability of the acyl-enzyme intermediate; (vi) validate that the designed Ser-His-Asp triad geometry stays within RMSD < 0.5 Å of the theozyme over a 1 μs trajectory. GROMACS

(e) The integrated stack. Co-folding (Boltz-2/BoltzGen) → catalytic ensemble (PLACER) → MLIP-MD (MACE-OFF24 + LES electrostatics) → QM/MM (GROMACS-CP2K / GENESIS-QSimulate-QM) → ABFE (Boltz-ABFE / OpenFE) — all chained in a single Nextflow DAG.

PART 3 — Polymer Informatics at the Frontier (Engine A)

(a) Property prediction beyond polyBERT. Stack: **polyBERT** (Kuenneth & Ramprasad, Nat. Commun. 14, 2023) as the 100M-PSMILES baseline; **TransPolymer** for comparison; **polyBART** (Agarwal et al., arXiv 2506.04233, Jun 2025) as the first bidirectional polymer language model (structure ↔ property, with experimentally synthesized validation). For the five canonical properties (Tg, Tc, density, FFV, Rg), the NeurIPS 2025 Open Polymer Prediction Challenge winning solution (James Day) showed **ModernBERT-base + property-specific ensembles + LAMMPS-derived 3D descriptors** beat both ChemBERTa and polyBERT — so the ceiling is an ensemble. For polymer foundation models: "Transferring a molecular foundation model for polymer property predictions" (Yang et al., arXiv 2310.16958) shows that BERT pretrained on Enamine REAL transfers competitively to polymers. For graph-based polymers, **PolymerGNN/MorganMG-style** approaches plus the **Polymer Genome platform** remain in use. Use **MoLFormer-XL** (Ross et al., IBM, 1.1B SMILES, linear attention + RoPE) for additional Engine A featurization. Hunter Heidenreich

(b) Generative polymer design. polyBART is the first true generative polymer LLM (PSELFIES tokenization, decoder-based generation); previous models were unidirectional property predictors. Use polyBART for Engine A's inverse design of segmented PU-urea — generate candidates with target Tg (low for stretch), elongation-at-break, *and* a designated carbamate cleavage site. Stack with multi-objective Bayesian optimization via **Atlas** + **Honegumi** + **Gryffin** for categorical (block sequence) + continuous (block ratio) variables. The frontier is to embed Engine B's carbamate-substrate compatibility as a constraint in the BO loop — i.e., the polymer fragment containing the carbamate must dock favorably to the candidate Engine B enzyme.

(c) Physics-based polymer validation. All-atom MD via GROMACS/OpenMM with **OPLS-AA/M** or **GAFF2** (validated against TransPolymer benchmarks); MACE-OFF24 for accurate side-chain interactions in elastane segments. Coarse-grained MD: **MARTINI 3** or **DPD** for microphase separation in segmented PU-urea. **SCFT** (self-consistent field theory) for predicting hard/soft block morphology — the soft-segment Tg below 0 °C and hard-segment crystallinity drive elastane stretch. LAMMPS for high-throughput modulus/elongation prediction (the same tool the NeurIPS 2025 winner used).

(d) High-throughput screening at maximum scale. Use **RDKit** enumeration for the segmented PU-urea chemical space (combinatorial enumeration of diisocyanate × diol × chain-extender backbones) → polyBERT/polyBART/ModernBERT embeddings → XGBoost PU multi-output models for the five polymer properties → MACE-OFF24-driven all-atom MD for the top 1% → LAMMPS modulus/elongation for the top 0.1% → enzyme-cleavability check via Boltz-2 docking against candidate enzyme. Target ceiling throughput: ~10^7 candidates enumerated, ~10^5 ML-scored, ~10^3 MD-validated, ~10^2 LAMMPS-validated, ~10 final passing both stretch + enzyme cleavability.

PART 4 — Data, MLOps, Orchestration & Infrastructure at the Frontier

(a) Workflow engines. The ceiling is **Nextflow / nf-core** (used by seqeralabs/nf-proteindesign for BoltzGen + ProteinMPNN + Boltz-2 + IPSAE pipelines as of late 2025) for the protein side, **Flyte** (Lyft / Union.ai) or **Prefect** for the polymer/MLOps side, with **Covalent** for hybrid HPC-cloud GPU dispatch. **Snakemake** for reproducible bioinformatics. The integrated DAG: RDKit enumeration → polyBERT/polyBART/XGBoost (Engine A) → theozyme extraction → RFdiffusion2/3 (Engine B) → EnhancedMPNN → Boltz-2 → PLACER → MACE-OFF24 MD → QM/MM (CP2K) → ABFE → Engine C passport generation.

(b) Experiment tracking + model registry. Weights & Biases + **MLflow** + **DVC** for data versioning + **Pachyderm** for reproducible data lineage. **Tamarind Bio** for hosted GPU pipeline orchestration (AlphaFold/RFdiffusion/MPNN/GROMACS exposed via API). Containerization via Docker/Singularity/Apptainer + **NVIDIA NGC** containers for CUDA-optimized molecular code. Containerized Conda environments for the polymer side. Tamarind

(c) Active learning / closed-loop / self-driving-lab orchestration. Atlas (Hickman et al., Digital Discovery 2025) as the BO brain; **Phoenics + Gryffin + Olympus + Golem** (Aspuru-Guzik group, originally 2018–2021 papers, still integrated in Atlas and ChemOS 2.0) as the algorithm library; **ChemOS 2.0** (Sim et al., Matter 2024, SiLA2-based) for hardware orchestration; **Honegumi** (Baird et al., arXiv 2502.06815, Feb 2025) for interactive BO script generation on Ax/BoTorch; **NIMO** (formerly NIMS-OS, Nov 2025) for legacy-hardware integration; **Bluesky** for multi-modal beamline experiments if any wet-lab synchrotron validation is involved. For purely in-silico active learning, use **BoTorch** + **GPyTorch**. Cell Press

(d) Scaling. Run RFdiffusion3 at batch sizes of 1000s on multi-GPU NVL72 racks; Boltz-2 inference at ~$0.001/prediction in 20s/GPU allows ~432,000 affinity predictions/day on a single H100, ~10^7 on a 24-GPU node. For MACE-OFF24 MD, 3×10^5 steps/day per A100 80GB scales to ~10^8 steps/day on a 32×GB200 NVL72 rack. Boltz-ABFE is the rate-limiter (FEP-class compute) — reserve for top ~100 designs. PubMed Central

(e) Frontier hardware. NVIDIA GB200 NVL72 rack (72× B200 + 36× Grace CPUs, **720 PFLOPS FP8 training / 1.44 exaFLOPS FP4 inference** per NVIDIA, ~$2-3M/rack) is the absolute on-prem ceiling. **B200** (180–192 GB HBM3e, 8 TB/s, $4.99–6.03/hr on RunPod). **H200** (141 GB, 4.8 TB/s, **$4.39/hr on RunPod** as of May 2026) is the price/performance sweet spot for Boltz-2/RFdiffusion. **AMD MI350X** as the licensing-diverse alternative. For specific-workload-tuned compute: **Cerebras CS-3** wafer-scale for training, **Groq LPU** for LLM agent latency, **Tenstorrent Wormhole** for budget alternatives. Data Center Dynamics SaaS Sentinel

PART 5 — Integration & the "Shouldn't Be Possible" Ceiling

(a) Tri-engine co-design loop. The frontier-of-the-frontier architecture is one in which Engine A's polymer chemistry, Engine B's enzyme, and Engine C's passport are NOT sequential but **jointly optimized**. Concretely: a single Atlas multi-objective BO over a vector (polymer backbone composition, enzyme sequence/scaffold, passport-relevant ESPR features) where each iteration:

1. Atlas proposes a (polymer, enzyme-template-scaffold) pair.
2. Engine A: RDKit + polyBART generate the polymer; polyBERT/polyBART/ModernBERT ensemble + XGBoost predict Tg, elongation, FFV, Rg, density; all-atom MACE-OFF24 MD verifies stretch.
3. The carbamate substrate is extracted (Engine A → B handoff).
4. Engine B: the substrate's TS geometry is computed (DFT or PLACER-prior); RFdiffusion2/3 generates backbones around the theozyme; EnhancedMPNN designs sequences; PLACER

validates multi-state preorganization; Boltz-2 scores binding + structure; MACE-OFF24 QM/MM verifies cleavage activation energy.

5. Engine C: Knowledge-graph + CatBoost classifier-chain + ChemBERTa-2/MoLFormer-XL featurization + LLM-RAG (Gemini 2.5 / Claude 4 with PaperQA2 grounding) generates the signed ESPR Digital Product Passport with GS1 QR.
6. Atlas updates surrogate, proposes next vector.

(b) Co-optimization specifics. The polymer's carbamate group must match the enzyme's active-site geometry. The frontier is to make this a **differentiable joint loss** — feed predicted enzyme kcat/KM (from Boltz-2 + PLACER scores) back into Atlas' acquisition function alongside polymer mechanical properties, weighted by ESPR recyclability scoring. This is genuinely novel and not published as of mid-2026 — it represents the "shouldn't be possible" ceiling.

(c) Agentic LLM orchestration — the "AI scientist" layer. Wrap the above DAG in a **multi-agent LLM system** modeled after **AI Scientist v2** (Sakana, arXiv 2504.08066) + **Google AI co-scientist** (Gemini 2.0) + **Robin** (FutureHouse) + **AgentD** (Ock et al., arXiv 2507.02925) + **ProtAgents** (Ghafarollahi & Buehler, Digital Discovery 2024) + **Kosmos** (Edison Scientific, Nov 2025; $70M seed Dec 2025 from Triatomic Capital, Spark Capital, with Jeff Dean and Dmitri Alperovitch as angels). Roles: a Planner agent (Gemini 2.5 / Claude 4 / GPT-5) reads UMG-SP2 literature via PaperQA2, proposes theozyme variants; a Designer agent invokes RFdiffusion2/3 via the AgentD pattern (tool calls); a Critic agent runs PLACER + Boltz-2 + QM/MM and writes a CRITIC report; an Engineer agent updates polymer chemistry in response to enzyme constraints; a Compliance agent assembles the Engine C passport. Kosmos-style runs at ~1,500 papers + 42,000 lines of analysis code per loop is the ceiling target. Tech Funding News Tech Funding News

(d) Per-engine absolute ceiling implementations.

- **Engine A ceiling**: RDKit (200K–1M candidates) → polyBERT + polyBART + ModernBERT-base ensemble (Tg / Tc / Rg / FFV / density) → MACE-OFF24 MD (top 0.1%) → LAMMPS modulus/elongation (top 0.01%) → Atlas BO with joint enzyme-cleavability constraint and MoLFormer-XL featurization for ChemBERTa-2 cross-validation.
- **Engine B ceiling**: UMG-SP2 transition-state theozyme (from Polêto et al. QM/MM TS5 coordinates, $\Delta G^\ddagger = 20.8$ kcal/mol) → RFdiffusion3 atom-level backbones (10K–100K) → EnhancedMPNN sequences (10× per backbone, ResiDPO-tuned) → PLACER multi-state preorganization filter (apo + Michaelis + tetrahedral TS + acyl-enzyme + deacylation TS) → Boltz-2 affinity + Boltz-ABFE absolute binding → MACE-OFF24 + LES MD validating intermediates → GROMACS 2026-CP2K QM/MM umbrella sampling for TS activation energy → FoldSeek novelty check ensuring TM-score < 0.5 to UMG-SP2 (and zero matches in UMG-SP1/SP2/SP3 cluster).

- **Engine C ceiling**: ChemBERTa-2 (Ahmad et al., arXiv 2209.01712) + MoLFormer-XL featurization of polymer + enzyme; CatBoost classifier-chain over ESPR regulatory predicates; SHACL knowledge graph for deterministic rule validation; Gemini 2.5 / Claude 4 LLM-RAG with PaperQA2 grounding; emit JSON-LD ESPR Digital Product Passport, sign with Ed25519, embed GS1 Digital Link in QR.

Recommendations

Phase 1 (validate the ceiling): Stand up the integrated Nextflow DAG on RunPod H200/B200 with RFdiffusion2 + LigandMPNN + Boltz-2 + PLACER for Engine B alone. Use the UMG-SP2 TS5 coordinates from Polêto et al. (RSC Chem. Sci. 2025) as the theozyme. Run 10,000 designs → expect 30–100 to pass PLACER preorganization + Boltz-2 + initial MACE-OFF24 MD. **Decision threshold**: if at least 5 designs have predicted $\Delta G^\ddagger \leqslant 18$ kcal/mol via QM/MM (CP2K) and TM-score < 0.5 to UMG-SP2 (FoldSeek), proceed to Phase 2.

Phase 2 (close the agentic loop): Add the Engine A polymer track (RDKit + polyBART + ModernBERT ensemble) and the Engine C passport track, then wrap the whole thing in an AgentD/ProtAgents-style LLM orchestrator using Atlas BO. Use Honegumi to generate the BO scripts. Run on a GB200 NVL72 if available (or on 8× H200 on RunPod at $4.39/hr/GPU as the floor). **Decision threshold**: full closed-loop runs converge within ⩽ 200 iterations to a (polymer, enzyme, passport) triple that simultaneously meets stretch (elongation ⩾ 400%, Tg ⩽ 0 °C), enzyme efficiency (predicted kcat/Km $\geqslant 3\times10^3\ M^{-1}s^{-1}$, $\Delta G^\ddagger \leqslant 18$ kcal/mol), and ESPR compliance.

Phase 3 (pure ceiling): Upgrade to RFdiffusion3 + EnhancedMPNN + Boltz-ABFE + MACELES-OFF; deploy a Kosmos-class LLM agent for autonomous hypothesis generation about novel diisocyanate chemistries; integrate Bluesky for any wet-lab synchrotron validation. **Decision threshold for declaring "ceiling reached"**: a designed enzyme experimentally validated with kcat/Km $\geqslant 10^4\ M^{-1}s^{-1}$ on the Engine A polymer-derived carbamate substrate, with FoldSeek novelty TM-score < 0.5 to any natural urethanase.

Licensing-safe stack (commercial-grade, MIT/Apache/CC-BY only):

- **AVOID**: AlphaFold3 (DeepMind non-commercial), Chai-2 (proprietary, early-access only), ESM3 (commercial restrictions).
- **USE**: RFdiffusion2 (MIT, github.com/RosettaCommons/RFdiffusion2), RFdiffusion3 (MIT, open-sourced Dec 3, 2025), LigandMPNN/EnhancedMPNN (MIT/Apache), Boltz-2/BoltzGen (MIT), PLACER (MIT, github.com/baker-laboratory/PLACER), FoldSeek (GPL-3 / MIT for binaries), MACE/MACE-OFF (MIT, github.com/ACEsuit/mace), OrbMol (Apache 2.0, huggingface.co/Orbital-Materials/OrbMol), UMA (CC-BY, huggingface.co/facebook/UMA), polyBERT (MIT), polyBART (MIT), GROMACS 2025/2026

(LGPL), OpenMM 8+ (MIT), LAMMPS (GPL), Atlas (MIT, github.com/aspuru-guzik-group/atlas), Honegumi (MIT), ChemOS 2.0 (MIT). uw

Caveats

1. **Boltz-2 affinity is not FEP**: independent benchmarks (Lukauskis et al. on molecular glues; Rowan Sci tracking dashboard at rowansci.com/blog/boltz2-benchmarks) show large variance by target class — molecular glues are particularly bad, with "generally poor or even negative correlations." Treat Boltz-2 as a fast ranker, not as a ground-truth FEP replacement. Reserve **Boltz-ABFE** or **OpenFE** for top candidates.
2. **RFdiffusion3 was open-sourced Dec 3, 2025** and its bioRxiv experimental validation is limited (35/190 cysteine hydrolases, no catalytically-dead mutants, no crystal structures of designed complexes — Engin Yapici / Medium reviewer notes this).
3. **PLACER does not replace QM/MM**: it scores active-site preorganization but does not directly compute activation energies — pair with GROMACS-CP2K or GENESIS-QSimulate-QM for the catalysis step.
4. **MACE-OFF24 has not been validated on every chemistry** — for carbamate hydrolysis TS specifically, anchor against DFT (e.g., ωB97M-V/def2-TZVPD) on a small QM region for benchmarking before scaling. Independent JACS 2025 reviewers note "slightly incorrect bond lengths and angles, incorrect stereochemistry at chiral centers and stereobonds and aromatic rings predicted to be non-planar" in some Boltz-2 outputs that propagate into downstream MD. arXiv
5. **Chai-2 / AlphaFold3 / ESM3 are licensing-blocked** for REWEAR-FUSED's commercial intent and must remain excluded.
6. **The "joint differentiable loss" across Engines A and B is novel and not yet published** — the Atlas BO surrogate approximation (handles mixed categorical/continuous) is the implementable substitute as of mid-2026.
7. **The 72-hour window is explicitly *not* a constraint** per the task definition — this is the no-limits ceiling spec, intended for a "shouldn't be possible" demonstration. Practical implementations may need to truncate at any phase boundary.
8. **Independent verification of Cradle/Adaptyv claims**: only 2 of 400 SPR-tested designs in the Adaptyv EGFR competition outperformed Cetuximab — the "frontier" antibody-design metrics in marketing materials often reflect tails of distributions and should be treated as illustrative, not guaranteed.